Bumperbot: `src` vs `src_latest` — Difference Report

Repository: `ChepuriNatraj/bumperbot` · Workspace: `/home/botforgelabs2/Desktop/bumperbot` · Generated by Hermes

> **Key finding — read this first.** `src_latest` is **not** a clean upgrade of `src`. It is missing **440 files** (almost all the AWS world `models/` assets needed for the `small_house`/`small_warehouse` Gazebo worlds) and, for several core files, it is the **older / buggy pre-fix version** — it even **reverts the laser-drift fix** and the Gazebo world-name fix that are present in `src`. On the other hand, `src_latest` *does* contain newer firmware/odometry calibration and a quaternion-based IMU pipeline. Do **not** blindly replace `src` with `src_latest`; cherry-pick only the improvements you want.

1. File-count comparison

624 files in `src` (current)

185 files in `src_latest`

440 removed in `src_latest`

1 added in `src_latest`

13 files changed (content)

Counts exclude junk (`.git`, `.vscode`, `__pycache__`, sqlite db files).

2. What was removed in `src_latest`

> **427 files** removed from `bumperbot_description/models/` — the entire AWS RoboMaker `small_house` & `small_warehouse` prop library (DAE meshes, PNG textures, `model.sdf`, `model.config`). Without these, the `small_house` / `small_warehouse` worlds load with **every prop missing** (this is exactly TROUBLESHOOTING.md issue #5). The `empty.world` still works.
> **13 files** removed from `rplidar_ros/sdk/` — duplicate macOS/Win32 SDK copies (e.g. `net_socket(1).cpp`). Harmless cleanup.

3. What was added in `src_latest`

> Exactly **1 real file**: `src_latest/src/resource/robot_firmware` — an ament resource-index marker (makes `robot_firmware` a properly-registered ament package). Minor.
> Everything else that *looked* new (`slam_bringup.launch.py`, `joystick_teleop.launch.py`, `twist_mux_*.yaml`, `robot_ros2_control.xacro`, `setup.py`, `firmware/imu/imu.ino`) **already exists in `src`** — they are not unique to `src_latest`.

4. Changed files (13) — what differs and which side is "newer"

File	What changed (src_latest vs src)	Direction
`robot_bringup/launch/bringup.launch.py`	`src` has the hardcoded `/home/amr/...` map & params paths repointed to this workspace (`/home/botforgelabs2/Desktop/bumperbot/...`). `src_latest` **still has the original `/home/amr/...` paths**, so it will fail to load the map/params on this machine. Timer offsets also differ slightly.	src newer (path fix)
`robot_bringup/config/nav2_params_default.yaml`	See first entry — whitespace + one `base_link` -> `base_footprint` for docking_server (present in both, rest is trailing-space only).	src newer (has fix)
`wheel_imu_localization/config/ekf.yaml`	**Critical:** `src` has `map_frame: odom` — the **laser-drift fix** I applied (EKF no longer fights AMCL for `map->odom`). `src_latest` still has `map_frame: map` — i.e. it **reverts the fix** and would reproduce the laser drift under Nav2.	src newer (drift fix)
`bumperbot_description/launch/robot_gazebo.launch.py`	`src` uses `.removesuffix('.world') + '.world'` — the documented fix for TROUBLESHOOTING #5 Symptom A (so `world_name:=small_house` resolves correctly). `src_latest` uses the simpler `world_name + '.world'` — the **buggy** form that triggered the failed Fuel download.	src newer (world fix)
`bumperbot_controller/launch/controller.launch.py`	`src` still launches the `twist_relay` node (needed so keyboard /cmd_vel reaches the DiffDriveController). `src_latest` **removed** that node — so in `src_latest`, plain teleop would not drive the robot unless twist_mux supplies the unstamped topic.	src safer (keeps relay)

bumperbot_controller/bumperbot_controller/twist_relay.py	`src` subscribes to `/cmd_vel` (works with plain teleop_twist_keyboard). `src_latest` subscribes to `/bumperbot_controller/cmd_vel_unstamped` — i.e. it was wired to the new `twist_mux` / joystick teleop pipeline (see joystick_teleop.launch.py).	src_latest newer (mux-aware)
bumperbot_description/urdf/robot.urdf.xacro	Geometry tweaks: `wheel_ygap` 0.027 -> 0.0075 (wheels pulled closer together), and the `imu_link` mount changed from `xyz=-0.082 0.01 ... rpy=0 0 pi` to `xyz=-0.082 -0.01 ... rpy=0 0 0` (flipped Y offset and upright orientation).	src_latest changed (geometry)
wheel_odometry/wheel_odometry/odom_nodef.py	Calibration: `wheel_radius` 0.0585 -> 0.0595, `wheel_base` 0.535 -> 0.540, `encoder_cpr` 46367 -> 46343.	src_latest newer (calib)
robot_firmware/robot_firmware/imu_serial.py	**Major rewrite (258 diff lines).** Rewritten to parse the new BNO055 quaternion CSV stream emitted by the updated imu.ino and publish `sensor_msgs/Imu`. Replaces the previous Euler-based parsing.	src_latest newer (rewrite)
robot_firmware/firmware/imu/imu.ino	**Major firmware improvement in src_latest.** Now reads the BNO055 native **quaternion** (`Q,w,x,y,z,gx,gy,gz,ax,ay,az`) instead of Euler angles — avoids gimbal-lock / REP-103 axis issues. Calibration status printed as `C,...` lines. Sign/axis correction is deferred to the Python side.	src_latest newer (quaternion)
robot_firmware/robot_firmware/motor_serial.py	Tuning: `max_linear` 0.4 -> 0.6 m/s, `max_angular` 0.4 -> 0.7 rad/s, `control_period` 0.01 -> 0.02 s, `wheel_base` 0.535 -> 0.540 m.	src_latest newer (tuning)
robot_firmware/firmware/motor_encoder/motor_encoder.ino	Calibration: `WHEEL_RADIUS` 0.0585 -> 0.0595 and `CPR` set to 46367 (matches the odometry node).	src_latest newer (calib)

5. Selected raw diffs (src → src_latest)

robot_bringup/launch/bringup.launch.py

```
@@ -73,9 +73,9 @@
         )
     ),
     launch_arguments={
```

```
- "map": "/home/botforgelabs2/Desktop/bumperbot/maps/my_map.yaml",
+ "map": "/home/amr/maps/my_map.yaml",
"use_sim_time": "false",
- "params_file":
"/home/botforgelabs2/Desktop/bumperbot/src/robot_bringup/config/nav2_params_default.yaml"
+ "params_file": "/home/amr/robot_ws/src/robot_bringup/config/nav2_params_default.yaml"
}.items()
)
@@ -89,7 +89,7 @@
),
launch_arguments={
"use_sim_time": "false",
- "params_file":
"/home/botforgelabs2/Desktop/bumperbot/src/robot_bringup/config/nav2_params_default.yaml"
+ "params_file": "/home/amr/robot_ws/src/robot_bringup/config/nav2_params_default.yaml"
}.items()
)
@@ -98,7 +98,7 @@
executable="rviz2",
arguments=[
"-d",
- "/opt/ros/jazzy/share/nav2_bringup/rviz/nav2_default_view.rviz"
+ "/home/amr/robot_ws/src/robot_bringup/config/nav2_default_view.rviz"
],
output="screen"
)
@@ -116,43 +116,43 @@
# 4 sec
TimerAction(
- period=4.0,
+ period=3.0,
actions=[motor]
),
# 7 sec
TimerAction(
- period=7.0,
+ period=4.0,
actions=[lidar]
),
# 11 sec
TimerAction(
- period=11.0,
+ period=5.0,
actions=[odom]
),
# 14 sec
TimerAction(
- period=14.0,
+ period=7.0,
actions=[ekf]
),
# 18 sec
TimerAction(
- period=18.0,
+ period=9.0,
actions=[localization]
),
- # 24 sec
+ #24 sec
TimerAction(
- period=24.0,
+ period=10.0,
actions=[navigation]
),
# 30 sec
TimerAction(
- period=30.0,
+ period=12.0,
actions=[rviz]
),
])
```

wheel_imu_localization/config/ekf.yaml

```
@@ -5,11 +5,7 @@
two_d_mode: true
publish_tf: true
publish_acceleration: false
- # EKF is odom-level ONLY: it publishes odom->base_footprint.
- # AMCL (in bringup.launch.py) is the sole publisher of map->odom.
```

```
- # Previously map_frame was set to 'map' here, so the EKF ALSO
- # broadcast map->odom and fought AMCL -> the laser "drifted" in RViz.
- map_frame: odom
+ map_frame: map
odom_frame: odom
base_link_frame: base_footprint
world_frame: odom
```

bumperbot_description/launch/robot_gazebo.launch.py

```
@@ -27,10 +27,7 @@
     world_path = PathJoinSubstitution([
     bumperbot_description,
     "worlds",
     - PythonExpression(expression=[
     - "'", LaunchConfiguration("world_name"), "'",
     - ".removesuffix('.world') + '.world'"
     - ])
     + PythonExpression(expression=["'", LaunchConfiguration("world_name"), "'", " + '.world'"])
     ]
     )
```

wheel_odometry/wheel_odometry/odom_nodef.py

```
@@ -33,17 +33,17 @@
     self.declare_parameter(
     "wheel_radius",
     - 0.0585
     + 0.0595
     )
     self.declare_parameter(
     "wheel_base",
     - 0.535
     + 0.540
     )
     self.declare_parameter(
     "encoder_cpr",
     - 46367 # can try with diff
     + 46343 # can try with diff
     )
     self.wheel_radius = (
```

robot_firmware/robot_firmware/motor_serial.py

```
@@ -19,13 +19,13 @@
     # ---- Parameters (tunable at launch, no code edits needed) ----
     self.declare_parameter("serial_port", "/dev/ttyACM1")
     self.declare_parameter("baud_rate", 115200)
     - self.declare_parameter("max_linear", 0.4) # m/s
     - self.declare_parameter("max_angular", 0.4) # rad/s
     + self.declare_parameter("max_linear", 0.6) # m/s
     + self.declare_parameter("max_angular", 0.7) # rad/s
     # max PWM change per cycle
     self.declare_parameter("cmd_timeout", 0.3) # seconds - stop if no /cmd_vel
     - self.declare_parameter("control_period", 0.01) #100 Hz
     -
     - self.declare_parameter("wheel_base", 0.535)
     + self.declare_parameter("control_period", 0.02)
     +
     + self.declare_parameter("wheel_base", 0.540)
     self.max_linear = self.get_parameter("max_linear").value
     self.max_angular = self.get_parameter("max_angular").value
```

6. How the workspace is structured

Both folders are standard ROS 2 **colcon** workspaces with one `src/` of packages:

Package	Role

`bumperbot_description`	URDF/Xacro model, Gazebo worlds, RViz, ros2_control HW interface, world `models/`
`bumperbot_controller`	ros2_control controllers (DiffDrive, simple/noisy velocity) + `twist_relay` + joystick/twist_mux teleop
`robot_firmware`	Real-hardware serial drivers (imu_serial, motor_serial, led_control) + Arduino .ino firmware
`wheel_odometry`	Encoders → `/wheel/odom`
`wheel_imu_localization`	EKF fusing wheel odom + IMU → `/odom`; Nav2 AMCL owns `map→odom`
`robot_bringup`	Top-level launch (bringup / slam_bringup) tying it together + Nav2 params
`rplidar_ros`	Vendored RPLiDAR A2 driver → `/scan`

TF chain: `odom→base_footprint` from EKF; `map→odom` from AMCL (Nav2). The laser is rigidly fixed to `laser_link→base_link→base_footprint`, so "laser drift" is always a transform-chain issue, never the LiDAR itself.

7. Recommendation

1. **Keep `src` as the working tree.** It has the laser-drift fix (`ekf.yaml map_frame: odom`), the base-frame unification, the `/home/amr`→workspace path fix, and the Gazebo `.removesuffix('.world')` fix — plus the full `models/` assets.

2. **If you want the firmware/IMU improvements from `src_latest`**, cherry-pick only these (they are safe and additive):

   - `robot_firmware/firmware/imu/imu.ino` + `robot_firmware/robot_firmware/imu_serial.py` (quaternion IMU)
   - `robot_firmware/robot_firmware/motor_serial.py`, `.../firmware/motor_encoder/motor_encoder.ino` (tuning)
   - `wheel_odometry/.../odom_nodef.py` (calibrated radius/base/cpr)
   - `bumperbot_description/urdf/robot.urdf.xacro` (geometry) — but re-test the IMU mount & wheel spacing

3. **Do NOT pull in** `src_latest`'s `ekf.yaml` (reverts drift fix), `bringup.launch.py`

> (restores broken paths), `robot_gazebo.launch.py` (restores world bug), or its stripped `models/`.

Note: `src` currently has one local commit (`bbf4ccd`) ahead of `origin/main` with the laser-drift fixes; `src_latest` is untracked and uncommitted.